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YUG AGARWAL

Backend Developer | AI, Computer Vision & Automation Enthusiast

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Professional Summary

Proactive third-year Computer Science (AI) student with experience in **backend development, automation, and system design**, contributing to **10+ real-world tools and systems**. Skilled in **Python, Java, SQL, DSA, and OOP**, with hands-on work in **Django, OpenCV, MediaPipe, MySQL, and Google Apps Script**. Built and deployed **8+ projects** including a **serverless CMS, AI-based automation systems, backend applications** and others, focusing on **scalability and performance**. Experience across **3+ roles in development, research, and operations**, with certifications in **Python, Data Analytics, and AI (NPTEL, CISCO, Infosys Springboard)**.

Work Experience

- **Backend Developer at Tranquiex, Ghaziabad** **(May 2025 – Sep 2025)**
  - *Developed 4+ core backend modules, led 20+ testing iterations, and proposed 15+ future feature enhancements.*
  - *Collaborated with a team of 6, explored 3+ new technologies, and supported project planning and team coordination.*
- **Research Analyst Intern at Nebulyn, Ghaziabad** **(Oct 2024 – Apr 2025)**
  - *Conducted 15+ detailed market and competitor analyses to uncover trends and support strategic planning.*
  - *Delivered 10+ insight-driven reports from multi-source data, aiding key business decisions.*
- **Operations Intern at Medipol Pharmaceuticals Pvt. Ltd., Noida** **(Aug 2023 – Sep 2023)**
  - *Completed a 5-week internship, assisting in 2+ major tender filings, order tracking, and process documentation.*
  - *Strengthened CorelDRAW skills by completing 8+ design tasks under expert guidance.*
- **Business Operations Support in Family Business** **(Apr 2023 – Aug 2023)**
  - *Created 50+ pricing spreadsheets with delivery, commissions, and distance-based calculations for e-commerce shift.*
  - *Handled 150+ product inventory and SKUs entries and optimized daily business data using advanced Excel features.*

Education

- **ABES Institute of Technology, Ghaziabad – 201009 (Pursuing)**
  Bachelors of Technology, Computer Science & Engineering, Specialization in Artificial Intelligence & Honors in Cyber Security
  *Abdul Kalam Technical University (AKTU – formerly UPTU) | CGPA: **7.51/10***
- **Bhai Parmanand Vidya Mandir School, Delhi – 110092**
  *Intermediate: PCM with Computer Science in CBSE Board | Cleared with **76.2%** in 2023*

Technical Skills

- **Programming Languages:** Python, Java, C
- **Core Concepts:** Data Structures & Algorithms, OOP, System Design Basics
- **Frameworks & Libraries:** Django, OpenCV, MediaPipe, NumPy, Pandas, Tkinter, Matplotlib, PyAutoGUI
- **Databases:** MySQL, SQLite, MongoDB
- **Web Development & Deployment:** HTML, CSS, JavaScript, website hosting, DNS & domain configuration, SEO optimization
- **Automation & Scripting:** PowerShell, Command Prompt (CMD), Google Apps Script, workflow automation
- **Tools & Technologies:** Git, VS Code, Netlify, Vercel, Google Analytics, Office 365 (Excel, Word, PowerPoint)
- **AI-Assisted Development:** ChatGPT, Claude, AntiGravity, DeepSeek (prototyping, debugging, automation workflows)

Projects

- **HubCenter – Serverless Notification Hub *(JavaScript, Netlify, Google workspace & automation tools)***
  *Built a **serverless CMS** using Google Sheets and Docs, with real-time caching, secure parsing, dynamic SEO, RSS feed, and **automated newsletter workflows**, enabling dynamic content management in static websites via **Google Apps Script**.*
- **HandScope – AI-Based System Control via Hand Gestures *(Python, MediaPipe, OpenCV, PyAutoGUI)***
  *Developed a hand gesture-controlled automation tool to perform **7+ system tasks** (e.g., volume control, mute/unmute, app launch, etc.) using real-time computer vision with over **90% accuracy**.*
- **RoomIQ – Intelligent Room-Based Energy Control *(Python, OpenCV, Raspberry Pi, Arduino, Firebase)***
  *Designed an AI-enabled classroom **automation** system that detects occupancy in **real time**, intelligently controls lights and fans, reduces unnecessary energy usage, and ensures smooth, autonomous operation.*
- **Telegram Bot for Billing & Stock Management *(Python, MySQL, Telegram API)***
  *Created a Telegram bot **handling 100+ billing** and inventory queries, integrating MySQL & Excel to streamline backend operations. Enhanced small business workflow **efficiency by 50%** through automated menu-driven chatbot interactions.*
- **MySQL Backup & Recovery Utility *(Python, MySQL)***
  *Developed a **1-click utility** for text-based MySQL backups with database/table selection; achieved **99%+ recovery** success in test runs, simplifying disaster recovery.*

- **Additional Automation, Utility and Management Projects**
  *Built a range of automation, utility and management tools including a **Late Comers Management System (Desktop & Web), Windows Login UI Replica, Cipher tools, system automation scripts**, and **database operation tools**, along with data processing tasks using **Pandas** and **NumPy**, showcasing strong problem-solving and rapid prototyping skills.*

Certifications

- Data Analytics with Python – NPTEL
- Data Science for Engineers – NPTEL
- The Joy of Computing using Python – NPTEL
- Python Essentials 1 – CISCO Networking Academy
- Database Management System 1 – Infosys Springboard
- Elements of AI – University of Helsinki (Minna Learn)

Achievements

- **Presenter – South Asian Conference on Science and Technology (SACST) 2026** ***(Mar 2026)***
  *Presented a research paper based on **RoomIQ**, an **AI-powered** classroom **energy automation system** using **edge-cloud architecture**, demonstrating real-time occupancy detection and **intelligent control of electrical devices**.*
- **Finalist – Code-A-thon 4.0 (3rd Year Mini Project Competition)** ***(Nov 2025)***
  *Reached the **Final Round** among **200+ teams** with **Project RoomIQ**, an AI-enabled smart classroom energy automation system.*
- **Internal Winner – Smart India Hackathon (SIH) 2025** ***(Sep 2025)***
  *Declared **internal round winner** among **200+ teams**, shortlisted in the **Top 50** teams for **PS ID: SIH25002**, and selected to represent the institute for national-level SIH.*
- **Finalist – Ideathon 3.0 (2nd Year Mini Project Competition)** ***(Jan 2025)***
  *Reached the **Final Round** among **200+ teams** with **Project HandScope**, an AI-based gesture-controlled automation system.*

Other Details

- **Position of Responsibility**
  - Elected **Student Coordinator, Institution's Innovation Council (IIC)** – Contributed to organizing **25+ innovation-driven activities** and fostering collaboration within a team of **30+ student members**.
  - **Registration Team Lead, Annual College Fest** – Led and managed a team of **12 members** to handle registrations for a fest with **3500+ footfall**, ensuring seamless operations and timely coordination.
  - **Organising Team Member, Hacknovate 7.0 (International Hackathon)** – Contributed to event planning and execution, managing team coordination and operations, while supporting sponsor outreach and onboarding.
- **Typing Speed:** 55 WPM+ with 98%+ Accuracy
- **Languages Known:** Hindi (Native Proficiency), English (Professional Working Proficiency)
- **Soft Skills:** Attention to detail, Communication, Problem Solving, Critical Thinking, Adaptability, Fast Learning, Teamwork & Leadership.